Midterm 1 ECON 695 Cheatsheet

$E[u_i] = \frac{1}{n}\sum_{i=1}^{n} \hat{u}_i$

$Y \sim$ random var; $\{Y_1, Y_2, Y_3\} \sim$ random sample

sample mean $\bar{Y}_n = \frac{1}{n}\sum_{i=1}^{n} Y_i$

sample variance $S_n^2 = \frac{1}{(n-1)}\sum_{i=1}^{n}(Y_i - \bar{Y}_n)^2$

Convergence in Probability

Z_n: sequence of rand var that converge in prob to b if for any $\varepsilon > 0$:

$\lim_{n\to\infty} P(|Z_n - b| < \varepsilon) = 1$ or $\text{plim}\, Z_n = b$

Weak Law of Large Numbers (WLLN)

If $\{Y_1, Y_2, \dots, Y_n\} \sim$ random sample from pop. with mean μ, variance σ^2 (both finite), then $\bar{Y} = \frac{1}{n}\sum_{i=1}^{n} Y_i$ (which is a random var) converges in probability to the population mean: $\text{plim}\, \bar{Y}_n = \mu$

$\Rightarrow \bar{Y}$ is a consistent estimator for μ

Properties of estimators

Unbiased Estimator	$E[\bar{Y}_n] = \mu$ (pop. mean)
Pop. variance	$Var[\bar{Y}_n] = Var\left[\frac{1}{n}\sum_{i=1}^{n} Y_i\right] = \frac{\sigma^2}{n}$
"d.f. corrected" sample var	$E[S_n^2] = \sigma^2$ (Proved in class)

Continuous Mapping Theorem

Suppose g is a continuous function defined on the same metric space as the random variable Z_n's. If $\text{plim}\, Z_n = b$ and g is continuous at b, we have: $\text{plim}\, g(Z_n) = g(b)$

Central Limit Theorem (CLT)

random sample: $\{Y_1, \dots, Y_n\}$ w/ pop. μ & σ^2. The distribution of the sample mean $\bar{Y}_n$ collapses to a normal distribution at the rate $\sqrt{n}$:

$\lim_{n\to\infty} P\left(\frac{\sqrt{n}(\bar{Y}_n - \mu)}{\sigma} \le z\right) = \Phi(z)$,

where Φ is cdf f/ standard normal $N(0,1)$ such that

$\text{plim}\, \frac{\sqrt{n}(\bar{Y}_n - \mu)}{\sigma} = N(0,1)$

Univariate Regression

$Y_i = \beta_0 + \beta_1 X_i + u$ where $(\beta_0, \beta_1) = \arg\min_{b_0, b_1} E[(Y_i - b_0 - b_1 X_i)^2]$ $a(\beta_0, \beta_1)$

FOC: $\beta_0 = E[Y_i] - \beta_1 E[X_i]$

$\beta_1 = \frac{Cov(Y_i, X_i)}{Var(X_i)}$

$E[u_i] = 0$

$E[u_i X_i] = 0$

Frisch-Waugh Theorem (FW)

$Y_i = \beta_1 + \beta_2 X_{2i} + \dots + \beta_k X_{ki} + \dots + \beta_K X_{Ki} + u_i$

We want to find β_k on X_{ki}?

$\beta_k = \frac{cov(Y_i, \tilde{\varepsilon}_i)}{var(\tilde{\varepsilon}_i)}$

$\tilde{\varepsilon}_i = X_{ki} - E^*[X_{ki} | X_{(-k)i}]$ is the residual over the projection of X_{ki} on "all" other covariates

- β_k = coefficient from univariate regression of Y_i on X_{ki}, after "partialling out" other X's.
- $E^*[\cdot]$ = best linear projection operator (to 'partial out' other regressors)
- $\tilde{\varepsilon}_i$ = residual from regressing X_{ki} on all other covariates.

Adding Non-orthogonal Covariate $\rightarrow$ yes OVB!

Y_i = log wage; Imm_i = immigration status $\begin{cases} 1 & \text{immigrant} \\ 0 & \text{if native} \end{cases}$ (dummy)

① $Y_i = \lambda_0 + \lambda_1 Imm_i + u_i$

② $Y_i = \beta_0 + \beta_1 Imm_i + \beta_2 Educ_i + \varepsilon_i$

not given: $Educ_i \perp Imm_i$

Since $\frac{1}{N}\sum Imm_i Educ_i$ is likely nonzero, how does $\hat{\lambda}_1$ relate to $\hat{\beta}_1$?

Sol: $\hat{\lambda}_1 = \hat{\beta}_1 + \hat{\beta}_2 \hat{\pi}_1 \Rightarrow$ bias of $\hat{\beta}_2 \hat{\pi}_1 > 0$

Proof: Consider aux. reg in sample:

$Educ_i = \hat{\pi}_0 + \hat{\pi}_1 Imm_i + \hat{\varepsilon}_i$

where $\frac{1}{N}\sum_{i=1}^{N} \hat{\varepsilon}_i = \frac{1}{N}\sum_{i=1}^{N} \hat{\varepsilon}_i Imm_i = 0$ $(\hat{\varepsilon}_i \perp Imm_i)$

We plug aux. reg into ②: $Y_i = \hat{\beta}_0 + \hat{\beta}_1 Imm_i + \hat{\beta}_2 Educ_i + \hat{\varepsilon}_i$

$= \hat{\beta}_0 + \hat{\beta}_1 Imm_i + \hat{\beta}_2(\hat{\pi}_0 + \hat{\pi}_1 Imm_i + \hat{\varepsilon}_i) + \hat{\varepsilon}_i$

$= (\hat{\beta}_0 + \hat{\beta}_2\hat{\pi}_0) + (\hat{\beta}_1 + \hat{\beta}_2\hat{\pi}_1) Imm_i + (\hat{\beta}_2 \hat{\varepsilon}_i + \hat{\varepsilon}_i)$

Since $\hat{\varepsilon}_i \perp (1, Imm_i)$, $\hat{\varepsilon}_i \perp (1, Imm_i, Educ_i)$:

$\frac{1}{N}\sum_{i=1}^{N} (\hat{\beta}_0 + \hat{\beta}_2\hat{\pi}_0)\cdot(\hat{\beta}_2 \hat{\varepsilon}_i + \hat{\varepsilon}_i) = 0$

$\frac{1}{N}\sum_{i=1}^{N} (\hat{\beta}_1 + \hat{\beta}_2\hat{\pi}_1) Imm_i \cdot (\hat{\beta}_2 \hat{\varepsilon}_i + \hat{\varepsilon}_i) = 0$

which suggest $\hat{\lambda}_0 = \hat{\beta}_0 + \hat{\beta}_2\hat{\pi}_0$, $\hat{\lambda}_1 = \hat{\beta}_1 + \hat{\beta}_2\hat{\pi}_1$ solve the FOC from ①:

$(\hat{\lambda}_0, \hat{\lambda}_1) = \arg\min_{a,b} \frac{1}{N}\sum_{i=1}^{N}(Y_i - a - b\, Imm_i)^2$

$\frac{\partial(\hat{\lambda}_0, \hat{\lambda}_1)}{\partial a} = 0 \Rightarrow \frac{1}{N}\sum_{i=1}^{N}(Y_i - \hat{\lambda}_0 - \hat{\lambda}_1 Imm_i) = 0$

$\frac{\partial(\hat{\lambda}_0, \hat{\lambda}_1)}{\partial b} = 0 \Rightarrow \frac{1}{N}\sum_{i=1}^{N}(Y_i - \hat{\lambda}_0 - \hat{\lambda}_1 Imm_i) \times Imm_i = 0$

$\Rightarrow \hat{\lambda}_1 = \hat{\beta}_1 + \hat{\beta}_2\hat{\pi}_1$, where OVB $= \hat{\beta}_2\hat{\pi}_1 > 0$

($Educ_i$ is positively correlated to log wage$_i$, and if immigrants are more educated than natives ($Imm_i = 0$), then $\hat{\lambda}_1 > \hat{\beta}_1$)

If we split Imm_i into 3 mutually exclusive groups:

	D_{1i}	D_{2i}	D_{3i}
(i) Asian Immigrants	1	0	1
(ii) Hispanic Immigrants	0	1	0
(iii) Other Immigrants	0	0	1

if $\vec{0}$ $\Rightarrow$ native

We can model this w/ single reg of logwage$_i$ on $(1, D_{1i}, D_{2i}, D_{3i})$

$Y_i = \beta_0 + \beta_1 D_{1i} + \beta_2 D_{2i} + \beta_3 D_{3i} + u_i$

where $u_i \perp (1, D_{1i}, D_{2i}, D_{3i})$

$\beta_0 = E[Y_i | \text{Natives}]$

$\beta_1 = E[Y_i | \text{Asian}] - E[Y_i | \text{Natives}]$

$\beta_2 = E[Y_i | \text{Hispanic}] - E[Y_i | \text{Natives}]$

$\beta_3 = E[Y_i | \text{Other Immigrant}] - E[Y_i | \text{Native}]$

$\hookrightarrow$ continued on back

From Review: random sample $\{Y_i, X_i\}_{i=1}^{n}$, $Y_i = \beta_0 + \beta_1 X_i$

$(\hat{\beta}_0, \hat{\beta}_1) = \arg\min_b \frac{1}{N}\sum_{i=1}^{N}(Y_i - b_0 - b_1 X_i)^2$

$\Rightarrow \hat{\beta}_0 = \bar{Y}_n - \hat{\beta}_1 \bar{X}_n$

$\Rightarrow \hat{\beta}_1 = \frac{\sum_{i=1}^{N}(Y_i - \bar{Y}) X_i}{\sum_{i=1}^{N}(X_i - \bar{X}) X_i} = \frac{\widehat{Cov}(X_i, Y_i)}{\widehat{Var}(X_i)}$ which uses the fact that

$\widehat{cov}(X_i, Y_i) = \sum_{i=1}^{N}(Y_i - \bar{Y})(X_i - \bar{X})$

$= \sum_{i=1}^{N}(Y_i - \bar{Y}) X_i$

$= \sum_{i=1}^{N}(X_i - \bar{X}) Y_i$

It is equivalent to write: $\hat{\beta}_1 = \frac{\sum_{i=1}^{N}(X_i - \bar{X}) Y_i}{\sum_{i=1}^{N}(X_i - \bar{X}) X_i}$

plug $Y_i = \beta_0 + \beta_1 X_i + u_i$:

$\hat{\beta}_1 = \frac{\sum_{i=1}^{N}(X_i - \bar{X})(\beta_0 + \beta_1 X_i + u_i)}{\sum_{i=1}^{N}(X_i - \bar{X}) X_i}$

$= \beta_0(0) + \beta_1(1) + \frac{\sum_{i=1}^{N}(X_i - \bar{X}) u_i}{\sum_{i=1}^{N}(X_i - \bar{X}) X_i}$

$= \beta_1 + \frac{\sum_{i=1}^{N}(X_i - \bar{X}) u_i}{\sum_{i=1}^{N}(X_i - \bar{X}) X_i}$

Without a constant: $Y_i = \beta X_i + u_i$, where

$\hat{\beta} = \arg\min_b \frac{1}{N}\sum_{i=1}^{N}(Y_i - X_i b)^2$. We get the FOC:

$\frac{\partial \frac{1}{N}\sum_{i=1}^{N}(Y_i - X_i b)^2}{\partial b} = -2\frac{1}{N}\sum_{i=1}^{N} X_i(Y_i - X_i b) = 0$

$\Rightarrow \hat{\beta} = \frac{\frac{1}{N}\sum_{i=1}^{N} X_i Y_i}{\frac{1}{N}\sum_{i=1}^{N} X_i^2}$

* Without controlling f/ a constant, the numerator is $\frac{1}{N}\sum_{i=1}^{N} X_i Y_i$ not the sample cov: $\widehat{cov}(X_i, Y_i) = \frac{1}{N}\sum_{i=1}^{N} X_i Y_i$ unless $\bar{X}\bar{Y} = 0$

Adding Orthogonal Covariate $\rightarrow$ no OVB!

(From PS3, 3: meant to show how adding a orthogonal covariate doesn't bias coefficients $\Rightarrow$ no OVB)

Given: X_i & $Z_i \sim$ covariates of Y_i; assume $\bar{X} = 0$, $\bar{Z} = 0$

Model: $Y_i = a_0 + a_1 X_i + v_i$

If $X_i \perp Z_i \mid \frac{1}{N}\sum_{i=1}^{N} X_i Z_i = 0$, then if we estimate model:

$Y_i = b_0 + b_1 X_i + b_2 Z_i + e_i$

by OLS, we get $\hat{b}_0 = \hat{a}_0$ and $\hat{b}_1 = \hat{a}_1$

$\Rightarrow$ dropping Z_i would not lead to OVB

Proof: Write out the FOC for each and apply FW to get b_1 f/ long model

1) For $Y_i = a_0 + a_1 X_i + v_i$:

$(\hat{a}_0, \hat{a}_1) = \arg\min_{a_0, a_1} \frac{1}{N}\sum_{i=1}^{N}(Y_i - a_0 - a_1 X_i)^2$

① $\Rightarrow \frac{\partial(\hat{a}_0, \hat{a}_1)}{\partial a_0} = \frac{\partial}{\partial a_0}\left(\frac{1}{N}\sum_{i=1}^{N}(Y_i - a_0 - a_1 X_i)^2\right)$

$\Rightarrow \frac{\partial(\hat{a}_0, \hat{a}_1)}{\partial a_0} = 0$ so we solve and get

$\Rightarrow \hat{a}_0 = \bar{Y} - \hat{a}_1 \bar{X}$ $\quad \bar{X} = 0$

$\Rightarrow \hat{a}_0 = \bar{Y}$ as $\bar{X} = 0$

② $\Rightarrow \frac{\partial(\hat{a}_0, \hat{a}_1)}{\partial a_0} = 0 \Rightarrow \hat{a}_1 = \frac{\widehat{cov}(Y_i, X_i)}{\widehat{Var}(X_i)}$

2) For $Y_i = b_0 + b_1 X_i + b_2 Z_i + e_i$:

$(\hat{b}_0, \hat{b}_1) = \arg\min_{b_0, b_1, b_2} \frac{1}{N}\sum_{i=1}^{N}(Y_i - b_0 - b_1 X_i - b_2 Z_i)^2$

$\Rightarrow \frac{\partial(\hat{b}_0, \hat{b}_1)}{\partial b_0} = 0 \Rightarrow \hat{b}_0 = \bar{Y} - \hat{b}_1\bar{X} - \hat{b}_2\bar{Z}$

$\Rightarrow \hat{b}_0 = \bar{Y}$ since $\bar{X}, \bar{Z} = 0$

We use FW to find $\hat{b}_1$ w/ auxillary reg of X_i on $(1, Z_i)$

$X_i = \hat{\pi}_0 + \hat{\pi}_1 Z_i + \hat{\varepsilon}_i$

aux reg $\sim$ univariate so:

$\hat{\pi}_1 = \frac{\widehat{cov}(X_i, Z_i)}{\widehat{var}(X_i)} = 0$

Since $\frac{1}{N}\sum_{i=1}^{N} X_i Z_i = 0$; $(X_i \perp Z_i)$

$\hat{\pi}_0 = \bar{X} - \bar{Z}\hat{\pi}_1 \Rightarrow \hat{\pi}_0 = 0$

$\Rightarrow \hat{\varepsilon}_i = X_i - \hat{\pi}_0 - \hat{\pi}_1 Z_i$

$\Rightarrow \hat{\varepsilon}_i = X_i$

By FW:

$\hat{b}_1 = \frac{\frac{1}{N}\sum_{i=1}^{N} Y_i \hat{\varepsilon}_i}{\frac{1}{N}\sum_{i=1}^{N} \hat{\varepsilon}_i^2}$ b/c $\hat{b}_1 = \frac{\frac{1}{N}\sum Y_i X_i}{\frac{1}{N}\sum X_i^2}$ and $X_i = \hat{\varepsilon}_i$

$= \frac{\frac{1}{N}\sum_{i=1}^{N} Y_i (X_i - \bar{X})}{\frac{1}{N}\sum_{i=1}^{N}(X_i - \bar{X})^2} = \frac{\widehat{cov}(X_i, Y_i)}{\widehat{var}(X_i)} = \hat{a}_1$

$\Rightarrow \hat{a}_0 = \hat{b}_0 = \bar{Y}$ and $\hat{b}_1 = \hat{a}_0$

$\Rightarrow$ ommitting orthogonal covariate $\neq$ OVB

(continued)

Finding Logwage gaps between Immigrant Groups

If we want to find the logwage gaps between each immigrant group & natives that can be explained by diff. in educ, we can consider a seperate reg of logwage on educ and a constant focused on Asian immigrants:

(a) Asian: $y_i = \beta_0^a + \beta_1^b \text{ Educ}_i + \epsilon_i$

(b) Natives: $y_i = \beta_0^b + \beta_1^b \text{ Educ}_i + \epsilon_i$

Then the diff. in mean logwage can be decomposed:

$$\bar{y}^a - \bar{y}^b = \left(\hat{\beta}_0^a + \hat{\beta}_1^a \overline{\text{Educ}}^a\right) - \left(\hat{\beta}_0^b + \hat{\beta}_1^b \overline{\text{Educ}}^b\right)$$

$$= \left(\hat{\beta}_0^a - \hat{\beta}_0^b\right) + \hat{\beta}_1^a \overline{\text{Educ}}^a - \hat{\beta}_1^b \overline{\text{Educ}}^b$$

$$= \underbrace{\left(\hat{\beta}_0^a - \hat{\beta}_0^b\right)}_{\text{unexplained by educ}} + \underbrace{\left(\hat{\beta}_1^a - \hat{\beta}_1^b\right) \overline{\text{Educ}}^a}_{\text{within}} + \underbrace{\hat{\beta}_1^b \left(\overline{\text{Educ}}^a - \overline{\text{Educ}}^b\right)}_{\substack{\text{between} \\ \text{(or explained by educ)}}}$$

$\Rightarrow$ The difference that can be explained by diff in educational attainment (composition effect) is

$$\hat{\beta}_1^b \left(\overline{\text{Educ}}^a - \overline{\text{Educ}}^b\right)$$

$\left(\hat{\beta}_0^a - \hat{\beta}_0^b\right)$	Captures baseline wage diff between groups not explained by educ
$\left(\hat{\beta}_1^a - \hat{\beta}_1^b\right) \overline{\text{Educ}}^a$	Measures how much of the wage gap is due to different returns to education (i.e even w/ same years of schooling, how differently education is rewarded between the groups).
$\hat{\beta}_1^b \left(\overline{\text{Educ}}^a - \overline{\text{Educ}}^b\right)$	Measures how much of the wage gap is due to differences in education levels themselves (i.e. b/c 1 group is more educated on average)